

MULTIPLE INTEGRALS

1 Evaluate:

(i) $\int_0^1 \int_0^2 xy \, dy \, dx$

Ans $\int_0^1 \int_0^2 xy \, dy \, dx$

$$= \int_0^1 x \left[\frac{y^2}{2} \right]_0^2 dx$$

$$= \int_0^1 x \cdot \frac{4}{2} dx$$

$$= 2 \left[\frac{x^2}{2} \right]_0^1$$

$$= 2 \times \frac{1}{2}$$

$$= \underline{\underline{1}}$$

(ii) $\int_0^\pi \int_0^\pi x \sin y \, dy \, dx$

Ans $\int_0^\pi \int_0^\pi x \sin y \, dy \, dx$

$$= \int_0^\pi x \left[-\cos y \right]_0^\pi dx$$

$$= \int_0^\pi x \left[-\cos \pi + \cos 0 \right] dx$$

$$= \int_0^\pi x \left[1 - \cos x \right] dx$$

$$= \int_0^\pi (x - x \cos x) dx$$

$$= \left[\frac{x^2}{2} - x \sin x - \cos x \right]_0^\pi$$

$$= \frac{\pi^2}{2} - \pi \sin \pi - \cos \pi + \cos 0$$

$$= \frac{\pi^2}{2} - 0 + 1 + 1$$

$$= \frac{\pi^2}{2} + 2$$

$$= \underline{\underline{\frac{\pi^2 + 4}{2}}}$$

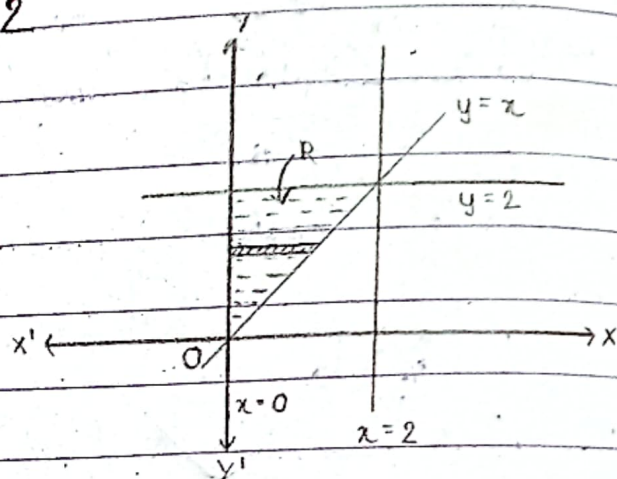
2. Sketch the region of integration and evaluate the following integrals by changing the order of integration

(i) $\int_0^2 \int_x^2 y^2 \sin xy \, dy \, dx$

Given region of integration is

$R: 0 \leq x \leq 2, \quad x \leq y \leq 2$

To reverse the order of integration, we consider horizontal strip for which the limits of x are $x=0$ to $x=y$ and such type of strip will then vary from $y=0$ to $y=2$.



By reversing the order of integration, the given region is redefined as

$R: 0 \leq x \leq y, \quad 0 \leq y \leq 2$

Now,

$$\int_0^2 \int_x^2 y^2 \sin xy \, dy \, dx = \int_0^2 \int_0^y y^2 \sin xy \, dx \, dy$$

$$= \int_0^2 y^2 \left[-\frac{1}{y} \cos xy \right]_0^y dy$$

$$= \int_0^2 y^2 \times \left(-\frac{1}{y} \right) (-\cos y^2 + \cos 0) dy$$

$$= \int_0^2 (y - y \cos y^2) dy$$

$$= \left[\frac{y^2}{2} - \frac{1}{2} \sin y^2 \right]_0^2$$

$$= \frac{1}{2} (4 - \sin 4 - 0 + \sin 0)$$

$$= \frac{4 - \sin 4}{2}$$

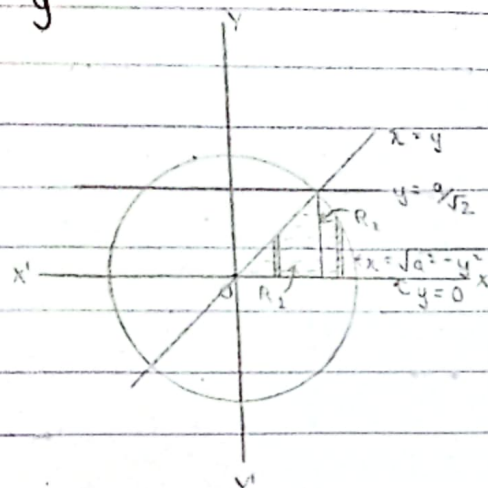
(ii) $\int_0^{a/\sqrt{2}} \int_y^{\sqrt{a^2-y^2}} x \, dx \, dy$

given region of integration is

$$R: 0 \leq y \leq a/\sqrt{2}, \quad y \leq x \leq \sqrt{a^2-y^2}$$

To reverse the order of integration, we consider two regions: R_1 and R_2 ; each having a vertical strip.

By reversing the order of integration, the given region is redefined as



For $R_1: 0 \leq x \leq a/\sqrt{2}, \quad 0 \leq y \leq x$

Now,

$$\iint_{R_1} f(x, y) \, dy \, dx = \int_0^{a/\sqrt{2}} \int_0^x x \, dy \, dx$$

$$= \int_0^{a/\sqrt{2}} x [y]_0^x \, dx$$

$$= \int_0^{a/\sqrt{2}} x^2 \, dx$$

$$= \left[\frac{x^3}{3} \right]_0^{a/\sqrt{2}}$$

$$= \frac{1}{3} \cdot \frac{a^3}{2\sqrt{2}}$$

$$= \frac{a^3}{6\sqrt{3}}$$

For $R_2: a/\sqrt{2} \leq x \leq a, \quad 0 \leq y \leq \sqrt{a^2-x^2}$

Now,

$$\iint_{R_2} f(x, y) \, dy \, dx = \int_{a/\sqrt{2}}^a \int_0^{\sqrt{a^2-x^2}} x \, dy \, dx$$

$$= \int_{a/\sqrt{2}}^a x [y]_0^{\sqrt{a^2-x^2}} dx$$

$$= \int_{a/\sqrt{2}}^a x \sqrt{a^2-x^2} dx$$

$$= -\frac{1}{2} \int_{a/\sqrt{2}}^a -2x(a^2-x^2)^{1/2} dx$$

$$= -\frac{1}{2} \left[\frac{(a^2-x^2)^{1/2+1}}{\frac{1}{2}+1} \right]_{a/\sqrt{2}}^a$$

$$= -\frac{1}{2} \times \frac{2}{3} \left[(a^2-x^2)^{3/2} \right]_{a/\sqrt{2}}^a$$

$$= -\frac{1}{3} \left[0 - (a^2/2)^{3/2} \right]$$

$$= a^3/6\sqrt{2}$$

Hence, for R

$$\iint_R f(x,y) dy dx = \iint_{R_1} f(x,y) dy dx + \iint_{R_2} f(x,y) dy dx$$

$$= a^3/6\sqrt{2} + a^3/6\sqrt{2}$$

$$= \frac{2a^3}{6\sqrt{2}}$$

$$= \frac{a^3\sqrt{2}}{6}$$

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(iii) $\int_0^{a/\sqrt{2}} \int_y^{\sqrt{a^2-y^2}} \log(x^2+y^2) dx dy, (a>0)$

Given region of integration is

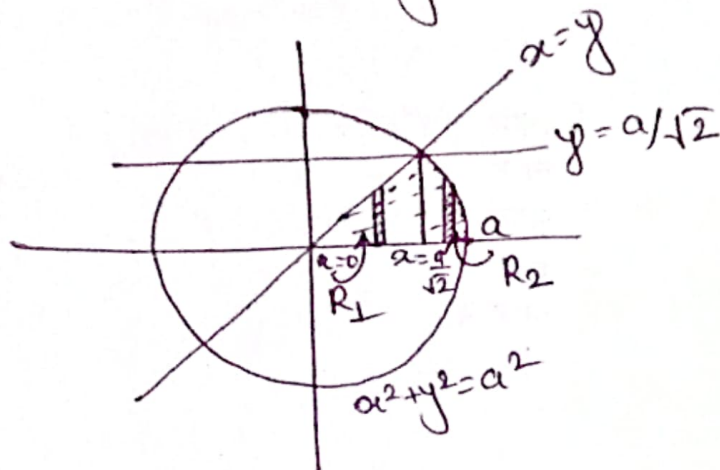
$$R: y \leq x \leq \sqrt{a^2-y^2}, \quad 0 \leq y \leq a/\sqrt{2}$$

Q.N. 2 (iii) $\int_0^{a/\sqrt{2}} \int_y^{\sqrt{a^2-y^2}} \log(x^2+y^2) dx dy, (a>0)$

Sol. 2

Here, the region of integration is,

$y \leq x \leq \sqrt{a^2-y^2}, 0 \leq y \leq \frac{a}{\sqrt{2}}$
as shown in figure below.



By changing order of integration, the region can be redefined as,

$$\left. \begin{array}{l} 0 \leq x \leq \frac{a}{\sqrt{2}} \\ 0 \leq y \leq x \end{array} \right\} \text{ for } R_1 \quad \& \quad \left. \begin{array}{l} \frac{a}{\sqrt{2}} \leq x \leq a \\ 0 \leq y \leq \sqrt{a^2-x^2} \end{array} \right\} \text{ for } R_2.$$

Now, for R_1 :

$$\begin{aligned} & \iint_{R_1} f(x,y) dy dx \\ &= \int_0^{a/\sqrt{2}} \int_0^x \log(x^2+y^2) dy dx \\ &= \int_0^{a/\sqrt{2}} \left[y \log(x^2+y^2) - 2y + 2x \tan^{-1} y/x \right]_0^x dx \\ &= \int_0^{a/\sqrt{2}} \left[x \log 2x^2 - 2x + 2x \cdot \frac{\pi}{4} - 0 \right] dx \end{aligned}$$

$$= \left[i \log_2 x^2 \int x - \int \left(\frac{d(i \log_2 x^2)}{dx} \int x dx \right) dx - \frac{x^2}{2} + \frac{\pi x^2}{4} \right]_0^{a/\sqrt{2}}$$

$$= \left[i \log_2 x^2 \cdot \frac{x^2}{2} - \int \frac{1}{2x^2} \cdot 4x \cdot \frac{x^2}{2} dx - x^2 + \frac{\pi x^2}{4} \right]_0^{a/\sqrt{2}}$$

$$= \left[\frac{x^2}{2} i \log_2 x^2 - \frac{x^2}{2} - x^2 + \frac{\pi x^2}{4} \right]_0^{a/\sqrt{2}}$$

$$= \frac{a^2}{4} i \log a^2 - \frac{3a^2}{4} + \frac{\pi a^2}{8} - 0$$

$$= \frac{a^2}{2} i \log a - \frac{3a^2}{4} + \frac{\pi a^2}{8} \text{ ——— ①}$$

For Re:-

$$\int_0^a \int_0^{\sqrt{a^2-x^2}} i \log(x^2+y^2) dy dx$$

$$= \int_0^a \left[y \log(x^2+y^2) - 2y + 2x \tan^{-1} y/x \right]_0^{\sqrt{a^2-x^2}} dx$$

$$= \int_0^a \left(\sqrt{a^2-x^2} i \log a^2 - 2\sqrt{a^2-x^2} + 2x \tan^{-1} \frac{\sqrt{a^2-x^2}}{x} - 0 \right) dx$$

$$= \int_0^a (i \log a^2 - 2) \sqrt{a^2-x^2} dx + \int_0^a 2x \tan^{-1} \frac{\sqrt{a^2-x^2}}{x} dx$$

$$= \cancel{\log(a^2 - 2)}$$

$$= (\log a^2 - 2) \int_{a/\sqrt{2}}^a \sqrt{a^2 - x^2} dx + 2 \int_{a/\sqrt{2}}^a x \tan^{-1} \frac{\sqrt{a^2 - x^2}}{x} dx$$

$$= (\log a^2 - 2) \left[\frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} \right]_{a/\sqrt{2}}^a + 2 \cdot \frac{a^2}{8}$$

$$= (\log a^2 - 2) \left[0 + \frac{a^2}{2} \cdot \frac{\pi}{2} - \frac{a^2}{4} - \frac{a^2}{2} \cdot \frac{\pi}{4} \right] + \frac{a^2}{4}$$

$$= (\log a^2 - 2) \left(\frac{\pi a^2}{8} - \frac{a^2}{4} \right) + \frac{a^2}{4}$$

$$= \frac{\pi a^2 \log a}{4} - \frac{a^2 \log a}{2} - \frac{\pi a^2}{4} + \frac{2a^2}{4} + \frac{a^2}{4}$$

$$= \frac{\pi a^2 \log a}{4} - \frac{a^2 \log a}{2} - \frac{\pi a^2}{4} + \frac{3a^2}{4}$$

— (ii)

So, from (i) & (ii),

$$\iint_R f(x,y) dy dx = \iint_{R_1} f(x,y) dy dx + \iint_{R_2} f(x,y) dy dx$$

$$= \frac{a^2}{2} \log a - \frac{3a^2}{4} + \frac{\pi a^2}{8} + \frac{\pi a^2 \log a}{4} - \frac{a^2 \log a}{2} - \frac{\pi a^2}{4} + \frac{3a^2}{4}$$

$$= \frac{\pi a^2 \log a}{4} - \frac{\pi a^2}{8}$$

$$= \frac{\pi a^2}{4} (\log a - 1/2) \quad \text{**}$$

Here, we have used two results directly,

$$\textcircled{1} \int \log(x^2+y^2) dy = y \log(x^2+y^2) - 2y + 2x \tan^{-1} \frac{y}{x}$$

proof

$$\text{LHS} = \int \log(x^2+y^2) dy$$

$$= \int 1 \cdot \log(x^2+y^2) dy$$

$$= y \log(x^2+y^2) - \int \frac{1}{x^2+y^2} \cdot 2y \cdot y dy$$

[Using formula of $\int u \cdot v dx$
where, $u = \log(x^2+y^2)$
 $v = 1$]

$$= y \log(x^2+y^2) - 2 \int \frac{y^2}{x^2+y^2} dy$$

$$= y \log(x^2+y^2) - 2 \int \left[1 - \frac{x^2}{x^2+y^2} \right] dy$$

$$= y \log(x^2+y^2) - 2 \left\{ y - x^2 \int \frac{dy}{x^2+y^2} \right\}$$

$$= y \log(x^2+y^2) - 2y + 2x^2 \cdot \frac{1}{x} \cdot \tan^{-1} \frac{y}{x}$$

$$= y \log(x^2+y^2) - 2y + 2x \tan^{-1} \frac{y}{x}$$

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$$\textcircled{9} \int_{a/\sqrt{2}}^a a \tan^{-1} \frac{\sqrt{a^2 - x^2}}{a} dx = \frac{a^2}{8}$$

put $x = a \sin \theta$, $dx = a \cos \theta d\theta$

when $x = a/\sqrt{2}$, $\theta = \pi/4$

when $x = a$, $\theta = \pi/2$.

$$\therefore \int_{\pi/4}^{\pi/2} a \sin \theta \cdot \tan^{-1} \frac{\sqrt{a^2 - a^2 \sin^2 \theta}}{a \sin \theta} \cdot a \cos \theta d\theta$$

$\rightarrow \tan^{-1}(\cot \theta)$

$$= \int_{\pi/4}^{\pi/2} a \sin \theta \tan^{-1} \left(\frac{a \cos \theta}{a \sin \theta} \right) \cdot a \cos \theta d\theta$$

$\rightarrow \tan^{-1}(\cot \theta) = \frac{\pi}{2} - \theta$

$$\begin{aligned} \tan \left(\frac{\pi}{2} - \theta \right) &= \cot \theta \\ \Rightarrow \frac{\pi}{2} - \theta &= \tan^{-1}(\cot \theta) \end{aligned}$$

$$= a^2 \int_{\pi/4}^{\pi/2} \sin \theta \cos \theta \left(\frac{\pi}{2} - \theta \right) d\theta$$

$$= \frac{a^2}{2} \int_{\pi/4}^{\pi/2} \sin 2\theta \left(\frac{\pi}{2} - \theta \right) d\theta$$

$$= \frac{a^2}{2} \int_{\pi/4}^{\pi/2} \sin 2\theta \left(\frac{\pi}{2} - \theta \right) d\theta$$

$$= \frac{\pi a^2}{4} \int_{\pi/4}^{\pi/2} \sin 2\theta d\theta - \frac{a^2}{2} \int_{\pi/4}^{\pi/2} \theta \sin 2\theta d\theta$$

$$= \frac{\pi a^2}{4} \left[-\frac{\cos 2\theta}{2} \right]_{\pi/4}^{\pi/2} - \frac{a^2}{2} \left[-\frac{\theta \cos 2\theta}{2} + \frac{\sin 2\theta}{4} \right]_{\pi/4}^{\pi/2}$$

$$= -\frac{\pi a^2}{8} [-1 - 0] - \frac{a^2}{2} \left[\frac{\pi}{4} + 0 - 0 - \frac{1}{4} \right]$$

$$= \frac{\pi a^2}{8} - \frac{\pi a^2}{8} + \frac{a^2}{8} = \frac{a^2}{8} \quad \#$$

3. Using the double integration, find the area of region bounded by the curves $y = \sin x$, the line $x = 0$ and $x = \pi/2$.

On the basis of the given curves and lines, the region of integration is as shown in figure below.

Now,

$$\text{Area} = \iint_R dy dx$$

$$= \int_0^{\pi/2} \int_0^{\sin x} dy dx$$

$$= \int_0^{\pi/2} [y]_0^{\sin x} dx$$

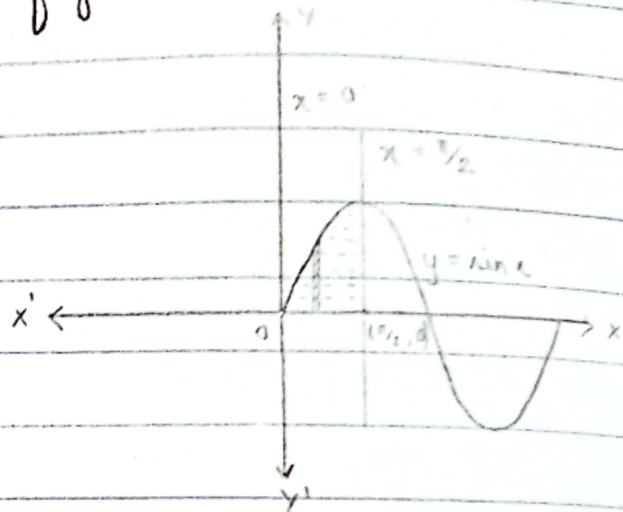
$$= \int_0^{\pi/2} \sin x dx$$

$$= [-\cos x]_0^{\pi/2}$$

$$= -\cos \pi/2 + \cos 0$$

$$= 0 + 1$$

$$= 1 \text{ sq. units}$$



4. Find the area bounded by the parabolas $x = y^2 - 1$ and $x = 2y^2 - 2$ by using double integration.

On the basis of the given parabolas, the region of integration is as shown in the figure.

Now,

$$\text{Area} = \iint_R dx dy$$

$$= \int_{-1}^1 \int_{2y^2-2}^{y^2-1} dx dy$$

$$= 2 \int_0^1 \int_{2y^2-2}^{y^2-1} dx dy$$

$$= 2 \int_0^1 [x]_{2y^2-2}^{y^2-1} dy$$

$$= 2 \int_0^1 (y^2-1-2y^2+2) dy$$

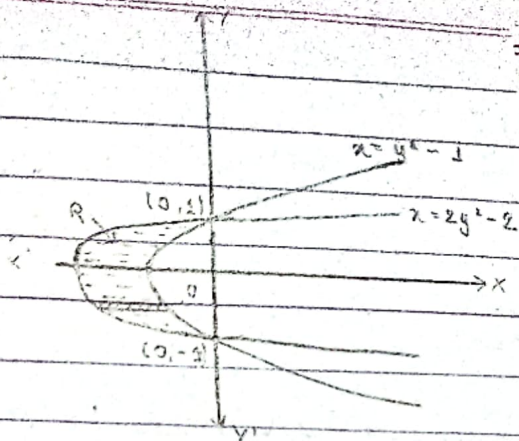
$$= 2 \int_0^1 (-y^2+1) dy$$

$$= 2 \left[-\frac{y^3}{3} + y \right]_0^1$$

$$= 2 \left[-\frac{1}{3} + 1 - 0 \right]$$

$$= 2 \times \frac{2}{3}$$

$$= \frac{4}{3} \text{ sq units}$$



5. Find the volume in the first octant bounded by coordinate plane, the cylinder $x^2 + y^2 = 4$ and the plane $z + y = 3$.

On the basis of the given functions, the region of integration is as shown in the figure.

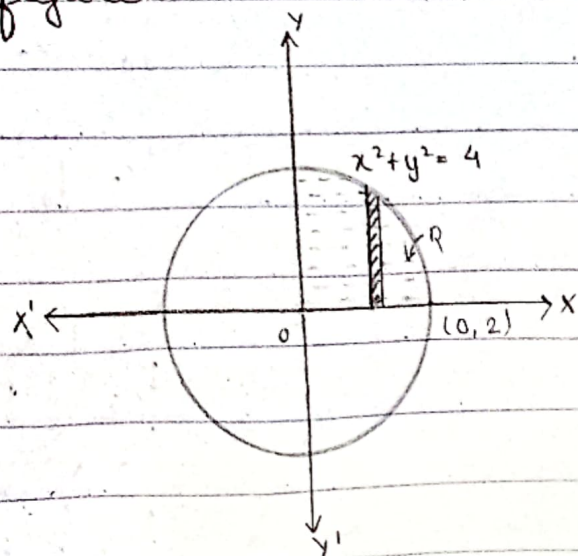
Now,

$$\text{Volume} = \iiint_R z \, dy \, dx$$

$$= \int_0^2 \int_0^{\sqrt{4-x^2}} (3-y) \, dy \, dx$$

$$= \int_0^2 \left[3y - \frac{y^2}{2} \right]_0^{\sqrt{4-x^2}} dx$$

$$= \int_0^2 \left[3\sqrt{4-x^2} - \frac{1}{2}(4-x^2) \right] dx$$



$$\begin{aligned}
 &= 3 \left[\frac{x}{2} \sqrt{4-x^2} + \frac{1}{2} \sin^{-1} \frac{x}{2} \right]_0^2 - \frac{1}{2} \left[4x - \frac{x^3}{3} \right]_0^2 \\
 &= 3 \left(\frac{2}{2} \sqrt{4-4} + \frac{1}{2} \sin^{-1} 1 - 0 \right) - \frac{1}{2} (8 - \frac{8}{3} - 0) \\
 &= 3 \times \frac{\pi}{2} - \frac{1}{2} \times \frac{16}{3} \\
 &= 3\pi - \frac{8}{3} \text{ cubic units}
 \end{aligned}$$

6. Find the volume of the tetrahedron bounded by the coordinate planes and the plane $x+y+z=1$

On the basis of the given functions, the region of integration is as shown in figure below.

Now,

$$\text{Volume} = \iint_R z \, dy \, dx$$

$$= \int_0^1 \int_0^{1-x} (1-x-y) \, dy \, dx$$

$$= \int_0^1 \left[(1-x)y - \frac{y^2}{2} \right]_0^{1-x} \, dx$$

$$= \int_0^1 \left[(1-x)^2 - \frac{1}{2} (1-x)^2 \right] \, dx$$

$$= \int_0^1 \frac{1}{2} (1-x)^2 \, dx$$

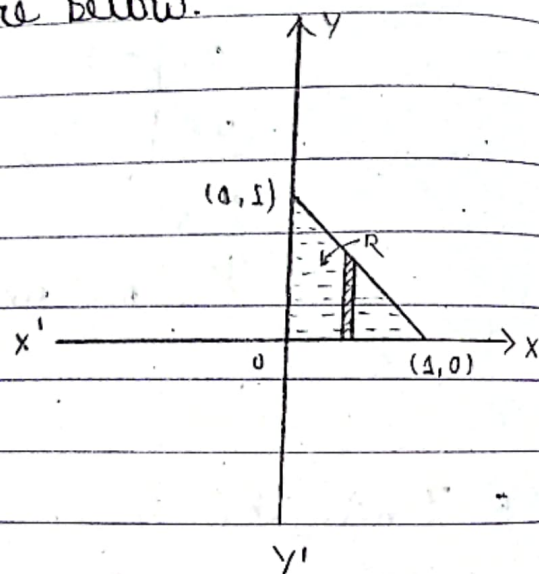
$$= \frac{1}{2} \int_0^1 (1-2x+x^2) \, dx$$

$$= \frac{1}{2} \left[x - 2 \times \frac{x^2}{2} + \frac{x^3}{3} \right]_0^1$$

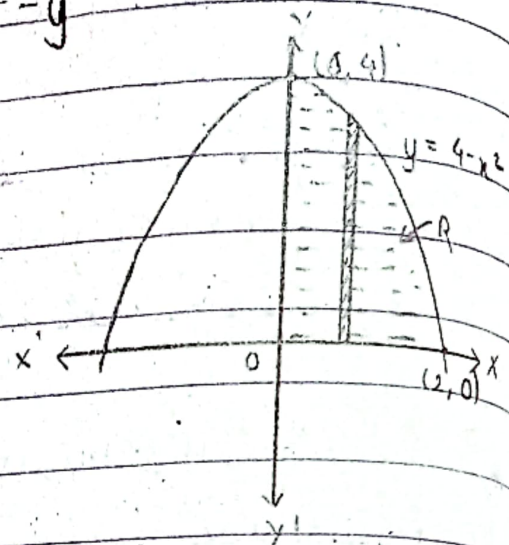
$$= \frac{1}{2} \left[1 - 1 + \frac{1}{3} - 0 \right]$$

$$= \frac{1}{2} \times \frac{1}{3}$$

$$= \frac{1}{6} \text{ cubic units}$$



8. Find the volume of the solid cut from the first octant by the surface $z = 4 - x^2 - y$.
 On the basis of the given functions, the region of integration is as shown



Now,

$$\text{Volume} = \iint_R z \, dy \, dx$$

$$= \int_0^2 \int_0^{4-x^2} (4-x^2-y) \, dy \, dx$$

$$= \int_0^2 \left[(4-x^2)y - \frac{y^2}{2} \right]_0^{4-x^2} dx$$

$$= \int_0^2 \left[(4-x^2)^2 - \frac{1}{2} (4-x^2)^2 \right] dx$$

$$= \int_0^2 \frac{1}{2} (4-x^2)^2 dx$$

$$= \frac{1}{2} \int_0^2 (16 - 8x^2 + x^4) dx$$

$$= \frac{1}{2} \left[16x - \frac{8}{3} x^3 + \frac{x^5}{5} \right]_0^2$$

$$= \frac{1}{2} (16 \times 2 - \frac{8}{3} \times 2^3 + \frac{2^5}{5} - 0)$$

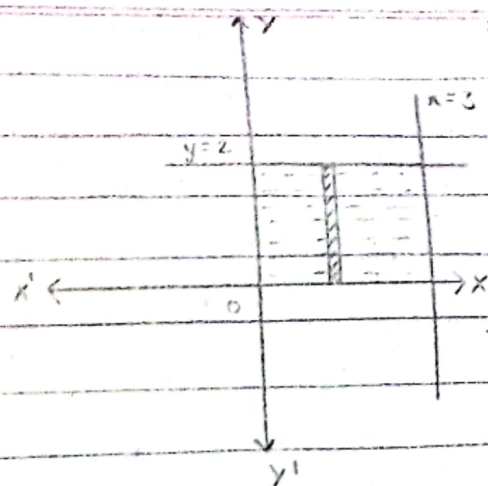
$$= \frac{1}{2} (32 - \frac{64}{3} + \frac{32}{5})$$

$$= \frac{1}{2} \times \frac{256}{15}$$

$$= \frac{128}{15} \text{ cubic units}$$

9. Find the volume of the solid in the first octant bounded by the coordinate planes, the plane $x = 3$ and the parabolic cylinder $z = 4 - y^2$.

On the basis of the given functions, the region of integration is as shown



Now,

$$\text{Volume} = \iint_R z \, dy \, dx$$

$$= \int_0^3 \int_0^2 (4-y^2) \, dy \, dx$$

$$= \int_0^3 \left[4y - \frac{y^3}{3} \right]_0^2 \, dx$$

$$= \int_0^3 (4 \times 2 - \frac{2^3}{3} - 0) \, dx$$

$$= \int_0^3 (8 - \frac{8}{3}) \, dx$$

$$= \frac{16}{3} \int_0^3 dx$$

$$= \frac{16}{3} [x]_0^3$$

$$= \frac{16}{3} (3-0)$$

$$= \frac{16}{3} \times 3$$

$$= 16 \text{ cubic units}$$

10. Evaluate by changing into polar integral.

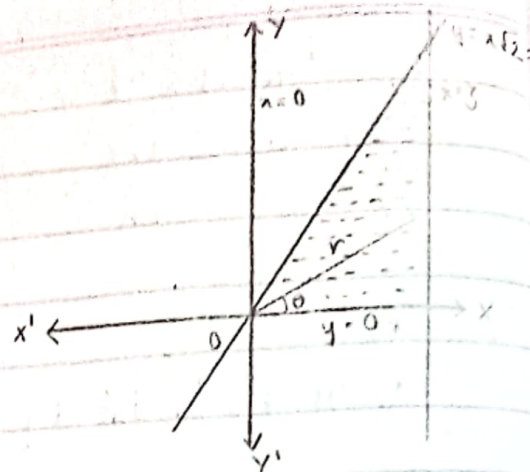
$$(i) \int_0^3 \int_0^{x\sqrt{2}} \frac{1}{\sqrt{x^2+y^2}} \, dy \, dx$$

Given region of integration is

$$0 \leq x \leq 3, \quad 0 \leq y \leq x\sqrt{2}$$

Let $x = r \cos \theta$, $y = r \sin \theta$, where
 $r^2 = x^2 + y^2$

Then, the region of integration
~~is~~ in polar coordinates is
 $0 \leq r \leq 3 \sec \theta$, $0 \leq \theta \leq \tan^{-1} \sqrt{2}$



Now, given integral becomes

$$\int_0^{\tan^{-1} \sqrt{2}} \int_0^{3 \sec \theta} \frac{1}{r} r dr d\theta$$

$$= \int_0^{\tan^{-1} \sqrt{2}} [r]_0^{3 \sec \theta} d\theta$$

$$= \int_0^{\tan^{-1} \sqrt{2}} 3 \sec \theta d\theta$$

$$\tan^2(\tan^{-1} \sqrt{2}) + 1$$

$$= \sec^2(\tan^{-1} \sqrt{2})$$

$$\text{or, } (\sqrt{2})^2 + 1 = \sec^2(\tan^{-1} \sqrt{2})$$

$$\text{or, } \sec \tan^{-1} \sqrt{2} = \pm \sqrt{3}$$

$$= 3 [\ln(\sec \theta + \tan \theta)]_0^{\tan^{-1} \sqrt{2}}$$

$$= 3 [\ln(\sec \tan^{-1} \sqrt{2} + \tan \tan^{-1} \sqrt{2}) - \ln(\sec 0 + \tan 0)]$$

$$= 3 [\ln(\sqrt{3} + \sqrt{2}) - \ln 1]$$

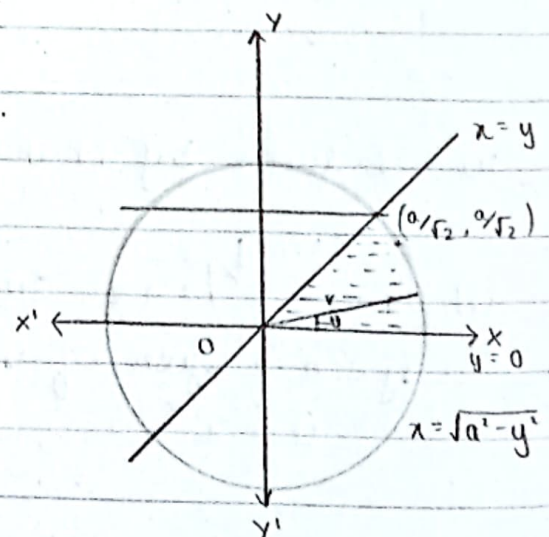
$$= 3 \ln(\sqrt{3} + \sqrt{2})$$

(ii) $\int_0^{a/\sqrt{2}} \int_y^{\sqrt{a^2 - y^2}} \log(x^2 + y^2) dx dy$, ($a > 0$)

given region of integration is
 $0 \leq y \leq a/\sqrt{2}$, $y \leq x \leq \sqrt{a^2 - y^2}$

Let $x = r \cos \theta$, $y = r \sin \theta$, where
 $r^2 = x^2 + y^2$

Then, the region of integration
in polar coordinates is
 $0 \leq \theta \leq \pi/4$, $0 \leq r \leq a$



Now, given integral becomes

$$\int_0^{\pi/4} \int_0^a \log r^2 r dr d\theta$$

$$= \int_0^{\pi/4} \int_0^a 2r \log r dr d\theta$$

$$= 2 \int_0^{\pi/4} \left[\log r \times \frac{r^2}{2} - \frac{1}{2} \times \frac{r^2}{2} \right]_0^a d\theta$$

$$= 2 \int_0^{\pi/4} \left(\frac{a^2}{2} \log a - \frac{a^2}{4} \right) d\theta$$

$$= 2 \times \frac{1}{2} \left(a^2 \log a - \frac{a^2}{2} \right) \int_0^{\pi/4} d\theta$$

$$= a^2 \left(\log a - \frac{1}{2} \right) \left[\theta \right]_0^{\pi/4}$$

$$= a^2 \left(\log a - \frac{1}{2} \right) \left(\frac{\pi}{4} - 0 \right)$$

$$= \frac{\pi a^2}{4} \left(\log a - \frac{1}{2} \right)$$

(iii) $\int_0^2 \int_0^{\sqrt{4-x^2}} \frac{xy}{\sqrt{x^2+y^2}} dy dx$

✓ given region of integration is
 $0 \leq x \leq 2, \quad 0 \leq y \leq \sqrt{4-x^2}$

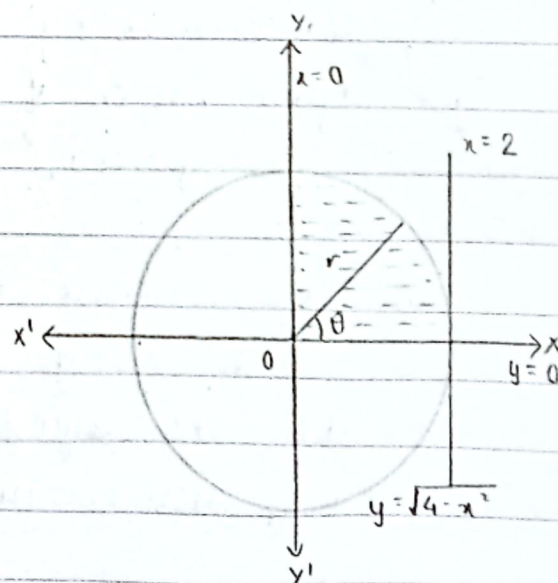
Let $x = r \cos \theta, \quad y = r \sin \theta$, where
 $r^2 = x^2 + y^2$

Then, the region of integration in polar coordinates is

$$0 \leq r \leq 2, \quad 0 \leq \theta \leq \pi/2$$

Now, given integral becomes

$$\int_0^{\pi/2} \int_0^2 r \cos \theta \cdot r \sin \theta \times \frac{1}{r} r dr d\theta$$



$$= \int_0^{\pi/2} \sin \theta \cos \theta \left[\frac{r^3}{3} \right]_0^2 d\theta$$

$$= \int_0^{\pi/2} \sin \theta \cos \theta \left(\frac{8}{3} - 0 \right) d\theta$$

$$= \frac{8}{3} \int_0^{\pi/2} \sin \theta \cos \theta d\theta$$

$$= \frac{8}{3} \times \frac{1}{2} \int_0^{\pi/2} 2 \sin \theta \cos \theta d\theta$$

$$= \frac{4}{3} \int_0^{\pi/2} \sin 2\theta d\theta$$

$$= \frac{4}{3} \left[-\frac{1}{2} \cos 2\theta \right]_0^{\pi/2}$$

$$= \frac{4}{3} \left(-\frac{1}{2} \cos \pi + \frac{1}{2} \cos 0 \right)$$

$$= \frac{4}{3} \left(\frac{1}{2} + \frac{1}{2} \right)$$

$$= \frac{4}{3} \times 1$$

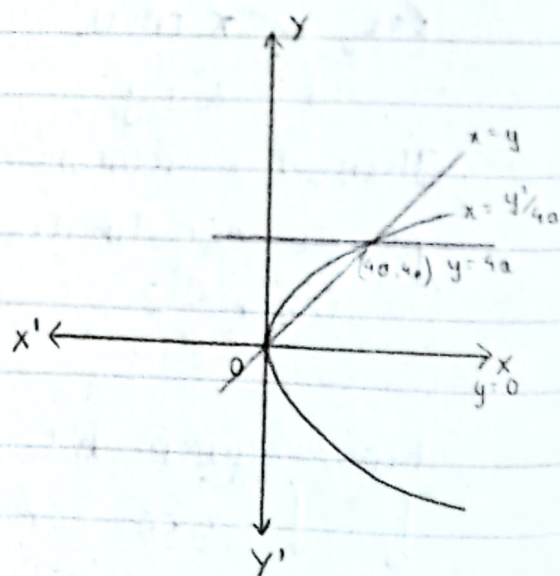
$$= \frac{4}{3}$$

$$(iv) \int_0^{4a} \int_{y^2/4a}^y \frac{x^2 - y^2}{x^2 + y^2} dx dy$$

given region of integration is
 $0 \leq y \leq 4a, \quad y^2/4a \leq x \leq y$

Let $x = r \cos \theta, y = r \sin \theta$, where
 $r^2 = x^2 + y^2$

Then, the region of integration in polar coordinates is



Example 22. Change the Cartesian Integral $\int_0^{4a} \int_y^{y^2/4a} \frac{x^2 - y^2}{x^2 + y^2} dx dy$ to polar form and integrate the polar integral.

Solution

We have the integral $\int_0^{4a} \int_y^{y^2/4a} \frac{x^2 - y^2}{x^2 + y^2} dx dy$

The region R of integration is bounded by the curves $x = y$, $x = \frac{y^2}{4a}$, i.e., $y^2 = 4ax$, $y = 0$, $y = 4a$, which is shown in figure.

In order to change into polar, put $x = r \cos\theta$, $y = r \sin\theta$ so that $x^2 + y^2 = r^2$ and $dx dy = r dr d\theta$.

For the limits of r : we have

$$y^2 = 4ax$$

$$\text{or, } (r \sin\theta)^2 = 4 ar \cos\theta$$

$$\text{or, } r^2 \sin^2\theta = 4 ar \cos\theta$$

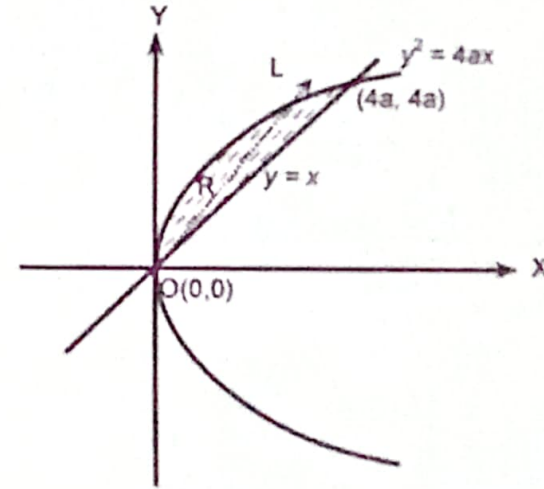
$$\therefore r = 0 \text{ and } r = \frac{4a \cos\theta}{\sin^2\theta} = 4a \cot\theta \operatorname{cosec}\theta$$

and the limits of θ are $\theta = \frac{\pi}{4}$ (slope of $y = x$) and $\theta = \frac{\pi}{2}$

(as the parabola touches y-axis).

Thus the polar equivalent of given Cartesian integral is,

$$\begin{aligned} \int_0^{4a} \int_y^{y^2/4a} \frac{x^2 - y^2}{x^2 + y^2} dx dy &= \int_{\pi/4}^{\pi/2} \int_0^{4a \cot\theta \operatorname{cosec}\theta} \frac{r^2(\cos^2\theta - \sin^2\theta)}{r^2} r dr d\theta \\ &= \int_{\pi/4}^{\pi/2} (\cos^2\theta - \sin^2\theta) \left[\frac{r^2}{2} \right]_0^{4a \cot\theta \operatorname{cosec}\theta} d\theta \end{aligned}$$



$$\begin{aligned}
&= \frac{1}{2} \int_{\pi/4}^{\pi/2} (\cos^2 \theta - \sin^2 \theta) 16a^2 (\cot^2 \theta \operatorname{cosec}^2 \theta - 0) \\
&= 8a^2 \int_{\pi/4}^{\pi/2} (\cot^4 \theta - \cot^2 \theta) d\theta \\
&= 8a^2 \int_{\pi/4}^{\pi/2} \cot^2 \theta (\cot^2 \theta - 1) d\theta \\
&= 8a^2 \int_{\pi/4}^{\pi/2} \cot^2 \theta (\operatorname{cosec}^2 \theta - 1 - 1) d\theta \\
&= 8a^2 \left[\int_{\pi/4}^{\pi/2} \cot^2 \theta \operatorname{cosec}^2 \theta d\theta - 2 \int_{\pi/4}^{\pi/2} \cot^2 \theta d\theta \right] \\
&= 8a^2 \left[- \int_{\pi/4}^{\pi/2} \cot^2 \theta d(\cot \theta) - 2 \int_{\pi/4}^{\pi/2} (\operatorname{cosec}^2 \theta - 1) d\theta \right] \\
&= -8a^2 \left[\left\{ \frac{\cot^3 \theta}{3} \right\}_{\pi/4}^{\pi/2} + 2 \left\{ -\cot \theta - \theta \right\}_{\pi/4}^{\pi/2} \right] \\
&= -8a^2 \left[\frac{1}{3} (0 - 1) + 2 \left\{ \left(0 + \frac{\pi}{2} \right) - \left(1 + \frac{\pi}{4} \right) \right\} \right] \\
&= -8a^2 \left[-\frac{1}{3} + 2 \left\{ \frac{\pi}{2} - 1 - \frac{\pi}{4} \right\} \right] \\
&= 8a^2 \left[\frac{1}{3} - 2 \left(\frac{\pi}{4} - 1 \right) \right] \\
&= 8a^2 \left[\frac{1}{3} - \frac{\pi}{2} + 2 \right] \\
&= 8a^2 \left[\frac{7}{3} - \frac{\pi}{2} \right]
\end{aligned}$$

$$\therefore \int_0^{4a} \int_y^{y^2/4a} \frac{x^2 - y^2}{x^2 + y^2} dx dy = 8a^2 \left[\frac{7}{3} - \frac{\pi}{2} \right].$$

11 Define the double integral of a function of two variable over a bounded region in the coordinate region. Does the order of integration matter? How are the limit of integration determined? Give examples.

— Let $f(x, y)$ be a function of two variables defined on a rectangular region $R: a \leq x \leq b, c \leq y \leq d$

Let us partition the given region into elementary portion, say $R_1, R_2, \dots, R_n$ of areas $\delta A_1, \delta A_2, \dots, \delta A_n$. Then,

$$\delta A_k = \delta x_k \delta y_k$$

Let S be the solid formed under the given surface $f(x, y)$ and above the region R . Then the approximated volume of the solid S is

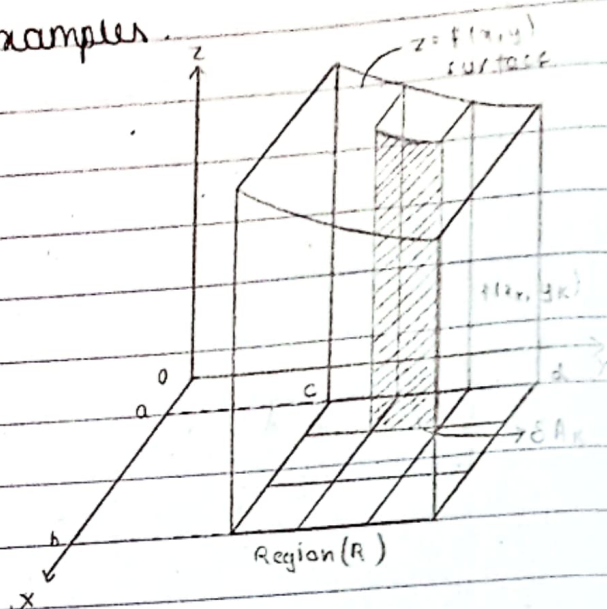
$$V \approx S_n = \sum_{k=1}^n f(x_k, y_k) \delta A_k$$

Then, a limit of S_n , when $n \rightarrow \infty$ (i.e. $\delta A_k \rightarrow 0$) is said to be double integral of $f(x, y)$ over R and is denoted by

$$\iint_R f(x, y) dA \text{ or } \iint_R f(x, y) dx dy$$

and is defined by

$$\iint_R f(x, y) dx dy = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k, y_k) \delta A_k$$



According to Fubini's Theorem,

$$\iint_R f(x, y) dA = \int_c^a \int_a^b f(x, y) dx dy = \int_a^b \int_c^d f(x, y) dy dx$$

Hence, The order of integration does not matter while performing double integration. However special attention must be paid to the limits while changing the order.

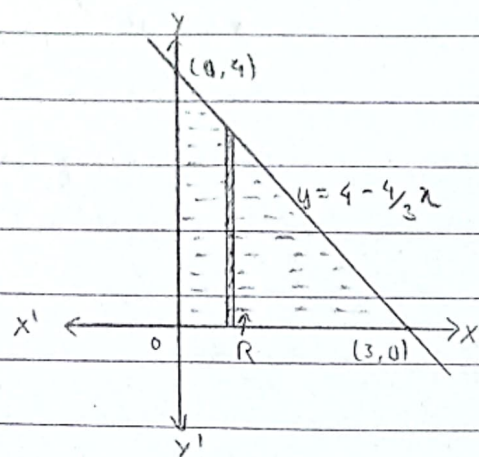
To determine the limit of an integration, the region of integration is first sketched based on the provided functions. Then, the inner limit of integration is determined with respect to outer variable. Similarly, The outer limit of integration is determined.

For example:

In the figure, y goes from x -axis to $y = 4 - \frac{4}{3}x$. Hence, the limits for y are $0 \leq y \leq 4 - \frac{4}{3}x$.

Also, x goes from 0 to 3. Hence, the limits for x are $0 \leq x \leq 3$.

Then, the integration can be written as



$$\iint_R f(x, y) dA = \int_0^3 \int_0^{4 - \frac{4}{3}x} f(x, y) dy dx$$

In this way, the limit of an integration is determined.

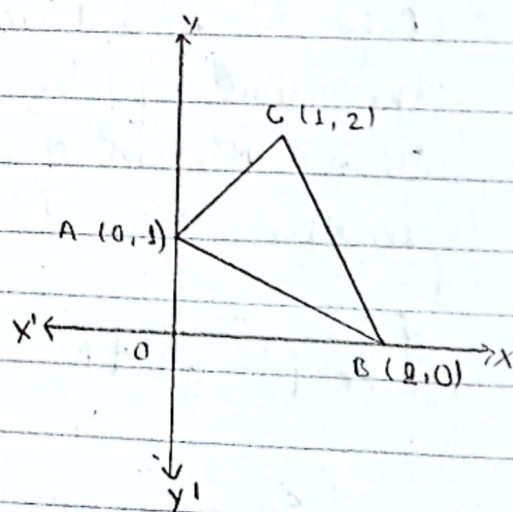
12. Why do we change the cartesian integral to polar integral?

How would you evaluate the double integral of a continuous function $f(x, y)$ over the region R in the xy -plane enclosed by the triangle with vertices $(0, 1)$, $(2, 0)$, and $(1, 2)$? give a reason for your answer.

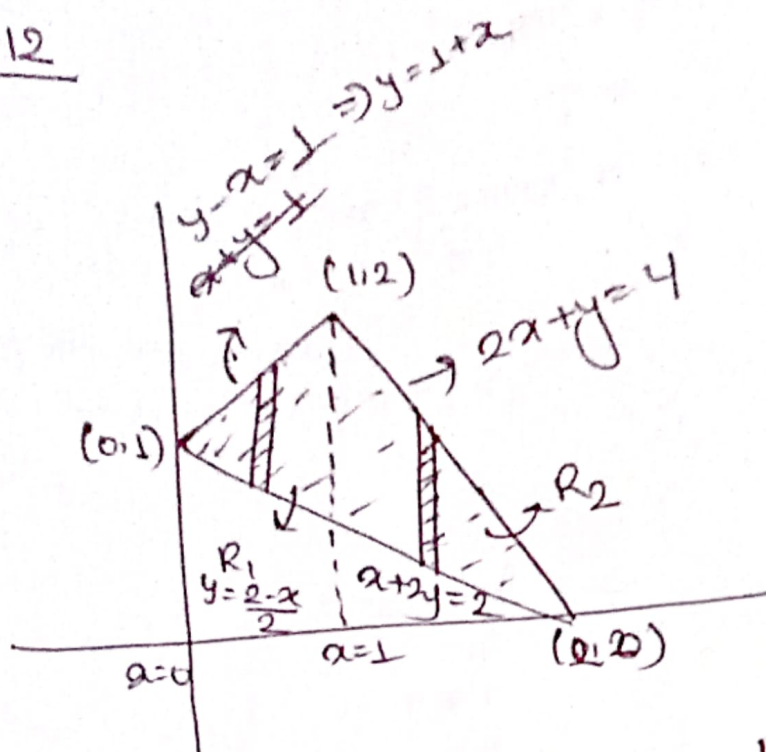
- In the transformation of cartesian form to polar form, the x is replaced by $r \cos \theta$, y by $r \sin \theta$ and $dx dy$ by $r dr d\theta$ with the limits of integration of r and θ

$$\text{i.e. } \int_{x_1}^{x_2} \int_{y_1}^{y_2} f(x, y) dy dx = \int_{\theta_1}^{\theta_2} \int_{r_1}^{r_2} f(r, \theta) r dr d\theta$$

The polar form is generally used when the region of integration is a circle or a part of it. This is because, in cartesian form, integrating over a circle would require setting up two regions and integrating them separately. However, in polar form these integrals are much easier to evaluate.



Q.N.12



Eq.n of the line passing through $(0,1)$ & $(1,2)$

$$y - y_1 = + \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$$

$$y - 1 = + \frac{2 - 1}{1 - 0} (x - 0)$$

$$y - 1 = + x$$

$$\boxed{y - x = 1}$$

Similarly for others as shown in figure.

Now,

$$\iint_R f(x,y) dy dx$$

$$= \iint_{R_1} f(x,y) dy dx + \iint_{R_2} f(x,y) dy dx$$

$$= \int_0^1 \int_{\frac{2-x}{2}}^{1+x} f(x,y) dy dx + \int_1^2 \int_{\frac{2-x}{2}}^{4-2x} f(x,y) dy dx$$

#

13. Evaluate the following triple integral

$$(i) \int_0^1 \int_0^2 \int_{x-z}^{x+z} (x+y+z) dy dx dz$$

$$\checkmark \int_0^1 \int_0^2 \int_{x-z}^{x+z} (x+y+z) dy dx dz$$

$$= \int_0^1 \int_0^2 \left[(x+z)y + \frac{y^2}{2} \right]_{x-z}^{x+z} dx dz$$

$$= \int_0^1 \int_0^2 \left[(x+z)(x+z) + \frac{1}{2}(x+z)^2 - (x+z)(x-z) - \frac{1}{2}(x-z)^2 \right] dx dz$$

$$= \int_0^1 \int_0^2 \left[\frac{3}{2}(x+z)^2 - (x^2 - z^2) - \frac{1}{2}(x-z)^2 \right] dx dz$$

$$= \frac{1}{2} \int_0^1 \int_0^2 (3x^2 + 6xz + 3z^2 - 2x^2 + 2z^2 - x^2 + 2xz - z^2) dx dz$$

$$= \frac{1}{2} \int_0^1 \int_0^2 (4z^2 + 8xz) dx dz$$

$$= \frac{1}{2} \times 2 \int_0^1 \int_0^2 (2z^2 + 4xz) dx dz$$

$$= \int_0^1 \left[2z^2 x + 4z \frac{x^2}{2} \right]_0^2 dz$$

$$= \int_0^1 (2z^3 + 2z^3) dz$$

$$= \int_0^1 4z^3 dz$$

$$= 4 \left[\frac{z^4}{4} \right]_0^1$$

$$= 4 \times \left(\frac{1}{4} - 0 \right)$$

$$= 4 \times \frac{1}{4}$$

$$= \underline{\underline{1}}$$

$$(ii) \int_1^e \int_1^{\log y} \int_1^{e^x} \log z \, dz \, dx \, dy$$

$$\int_1^e \int_1^{\log y} \int_1^{e^x} \log z \, dz \, dx \, dy$$

$$= \int_1^e \int_1^{\log y} [z \log z - z]_1^{e^x} dx \, dy$$

$$= \int_1^e \int_1^{\log y} [e^x \log e^x - e^x - \log 1 + 1] dx \, dy$$

$$= \int_1^e \int_1^{\log y} (xe^x - e^x + 1) dx \, dy$$

$$= \int_1^e [xe^x - e^x - e^x + x]_1^{\log y} dy$$

$$= \int_1^e [xe^x - 2e^x + x]_1^{\log y} dy$$

$$= \int_1^e [\log y e^{\log y} - 2e^{\log y} + \log y - e + 2e - 1] dy$$

$$= \int_1^e (y \log y - 2y + \log y + e - 1) dy$$

$$= [\log y \times y^2/2 - 1/2 \times y^2/2 - 2 \times y^2/2 + y \log y - y + ey - y]_1^e$$

$$= [1/2 y^2 \log y - 1/4 y^2 - y^2 + y \log y - 2y + ey]_1^e$$

$$= 1/2 e^2 \log e - 1/4 e^2 - e^2 + e \log e - 2e + e^2 - 1/2 \log 1 + 1/4$$

$$+ 1 - \log 1 + 2 - e$$

$$= 1/2 e^2 - 1/4 e^2 - e^2 + e \log e - 2e + e^2 - 0 + 1/4 + 1 - 0 + 2 - e$$

$$= 1/4 e^2 - 2e + 13/4$$

$$= 1/4 (e^2 - 8e + 13)$$

$$(iii) \int_0^{\pi/2} \int_0^{a \sin \theta} \int_0^{\frac{a^2-r^2}{a}} r dz dr d\theta$$

$$\int_0^{\pi/2} \int_0^{a \sin \theta} \int_0^{\frac{a^2-r^2}{a}} r dz dr d\theta$$

$$= \int_0^{\pi/2} \int_0^{a \sin \theta} r \left[z \right]_0^{\frac{a^2-r^2}{a}} dr d\theta$$

$$= \int_0^{\pi/2} \int_0^{a \sin \theta} r \left(\frac{a^2-r^2}{a} \right) dr d\theta$$

$$= \frac{1}{a} \int_0^{\pi/2} \int_0^{a \sin \theta} (a^2 r - r^3) dr d\theta$$

$$= \frac{1}{a} \int_0^{\pi/2} \left[a^2 \frac{r^2}{2} - \frac{r^4}{4} \right]_0^{a \sin \theta} d\theta$$

$$= \frac{1}{2a} \int_0^{\pi/2} (a^4 \sin^2 \theta - \frac{1}{2} a^4 \sin^4 \theta - 0) d\theta$$

$$= \frac{1}{2a^3} \int_0^{\pi/2} (\sin^2 \theta - \frac{1}{2} \sin^4 \theta) d\theta$$

$$= \frac{a^3}{2} \int_0^{\pi/2} \left[\frac{1}{2} (1 - \cos 2\theta) - \frac{1}{2} \left\{ \frac{1}{2} (1 - \cos 2\theta) \right\}^2 \right] d\theta$$

$$= \frac{a^3}{2} \times \frac{1}{2} \int_0^{\pi/2} \left[1 - \cos 2\theta - \frac{1}{4} (1 - 2\cos 2\theta + \cos^2 2\theta) \right] d\theta$$

$$= \frac{a^3}{4} \left[\theta - \frac{1}{2} \sin 2\theta - \frac{1}{4} \theta + \frac{1}{4} \times \frac{1}{2} \sin 2\theta - \frac{1}{4} \times \frac{1}{2} \left(\frac{1}{2} \sin 2\theta + \theta \right) \right]_0^{\pi/2}$$

$$= \frac{a^3}{4} \left[\theta - \frac{1}{2} \sin 2\theta - \frac{1}{4} \theta + \frac{1}{4} \sin 2\theta - \frac{1}{16} \sin 2\theta - \frac{1}{8} \theta \right]_0^{\pi/2}$$

$$= \frac{a^3}{4} \left[\frac{5}{8} \theta - \frac{5}{16} \sin 2\theta \right]_0^{\pi/2}$$

$$= \frac{a^3}{4} \left(\frac{5}{8} \times \frac{\pi}{2} - \frac{5}{16} \sin 2 \times \frac{\pi}{2} - 0 \right)$$

$$= \frac{a^3}{4} \times \frac{5\pi}{16}$$

$$= \frac{5a^3\pi}{64}$$

64

14. Evaluate $\iiint x^{l-1} y^{m-1} z^{n-1} dx dy dz$, where x, y, z are all positive but $(x/a)^p + (y/b)^q + (z/c)^r \leq 1$.

let

$$\begin{aligned} u &= (x/a)^p, & v &= (y/b)^q, & w &= (z/c)^r \\ x &= u^{1/p} a, & y &= v^{1/q} b, & z &= w^{1/r} c \\ dx &= a u^{1/p-1} du, & dy &= b v^{1/q-1} dv, & dz &= c w^{1/r-1} dw \end{aligned}$$

Now,

$$\begin{aligned} \iiint x^{l-1} y^{m-1} z^{n-1} dx dy dz &= \iiint (u^{1/p} a)^{l-1} (v^{1/q} b)^{m-1} (w^{1/r} c)^{n-1} \\ &\quad \left(a u^{1/p-1} du \right) \left(b v^{1/q-1} dv \right) \left(c w^{1/r-1} dw \right) \\ &= a^l b^m c^n \iiint u^{l/p-1} v^{m/q-1} w^{n/r-1} du dv dw \\ &\quad pqr \text{ where } (x/a)^p + (y/b)^q + (z/c)^r \leq 1 \end{aligned}$$

using Dirichlet's condition, we have

$$\iiint x^{l-1} y^{m-1} z^{n-1} dx dy dz = \frac{a^l b^m c^n}{pqr} \frac{\Gamma(1/p) \Gamma(1/q) \Gamma(1/r)}{\Gamma(1/p + 1/q + 1/r + 1)}$$

15. Find by triple integration, the volume of sphere $x^2 + y^2 + z^2 = a^2$

given sphere is

$$x^2 + y^2 + z^2 = a^2$$

The region of integration for first octant is

$$0 \leq x \leq a$$

$$0 \leq y \leq \sqrt{a^2 - x^2}$$

$$0 \leq z \leq \sqrt{a^2 - x^2 - y^2}$$

Now,

Volume = 8 × volume in first octant

$$= 8 \int_0^a \int_0^{\sqrt{a^2-x^2}} \int_0^{\sqrt{a^2-x^2-y^2}} dz dy dx$$

$$= 8 \int_0^a \int_0^{\sqrt{a^2-x^2}} [z]_0^{\sqrt{a^2-x^2-y^2}} dy dx$$

$$= 8 \int_0^a \int_0^{\sqrt{a^2-x^2}} \sqrt{a^2-x^2-y^2} dy dx$$

$$= 8 \int_0^a \left[\frac{y}{2} \sqrt{a^2-x^2-y^2} + \frac{1}{2} (a^2-x^2) \sin^{-1} \frac{y}{\sqrt{a^2-x^2}} \right]_0^{\sqrt{a^2-x^2}} dx$$

$$= 8 \int_0^a \left(\frac{1}{2} \sqrt{a^2-x^2} \sqrt{a^2-x^2-a^2} + \frac{1}{2} (a^2-x^2) \sin^{-1} 1 - 0 \right) dx$$

$$= 8 \times \frac{1}{2} \int_0^a (a^2-x^2)^{1/2} dx$$

$$= 4 \times \frac{\pi}{2} \left[a^2 x - \frac{x^3}{3} \right]_0^a$$

$$= 2\pi (a^2 - \frac{a^3}{3} - 0)$$

$$= 2\pi \times \frac{2a^3}{3}$$

$$= \frac{4\pi a^3}{3} \text{ cubic units}$$

Sieel
08/24